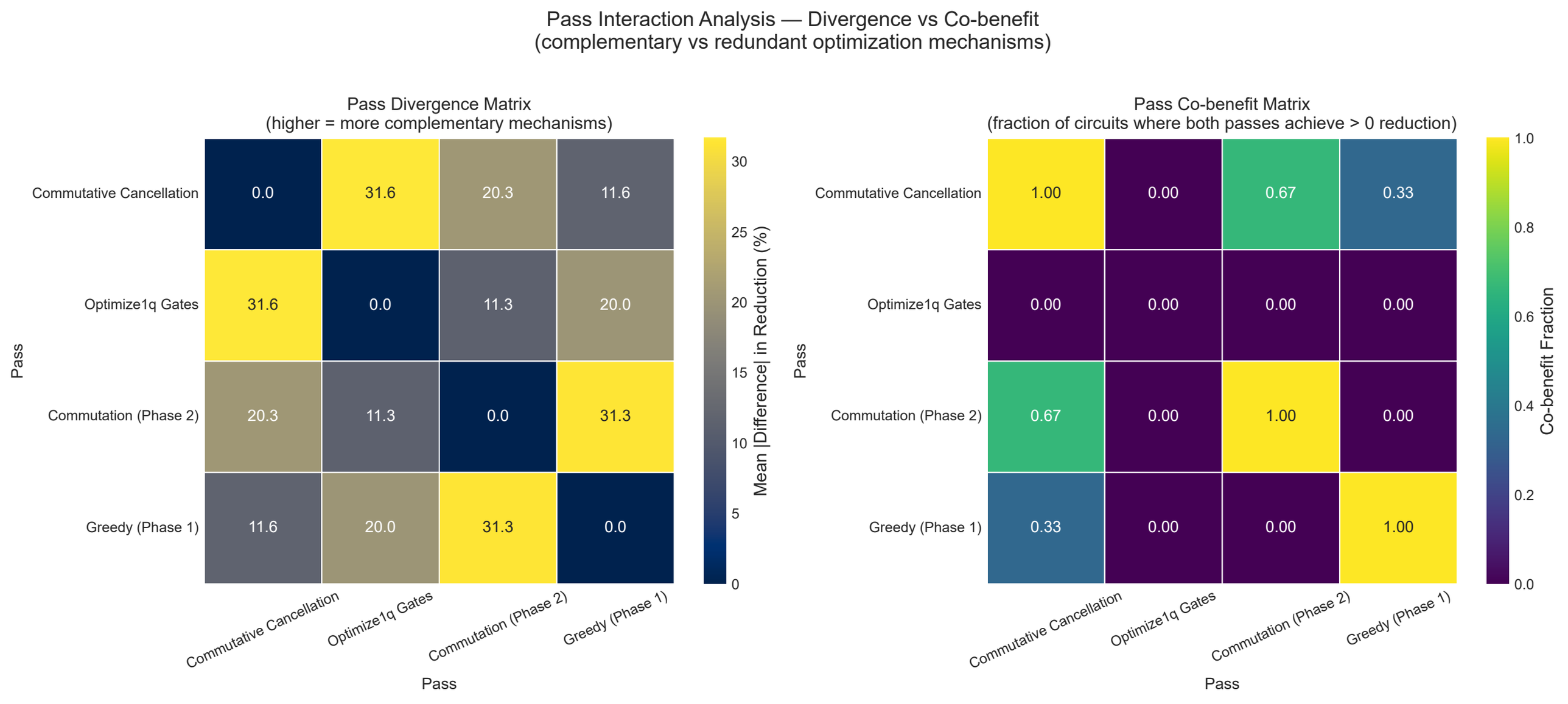
Pass Interaction Analysis — Divergence vs Co-benefit
(complementary vs redundant optimization mechanisms)
Pass Divergence Matrix
(higher = more complementary mechanisms)
Commutative Cancellation
Optimize1q Gates
Commutation (Phase 2)
Greedy (Phase 1)
0.0
31.6
20.3
11.6
31.6
0.0
11.3
20.0
20.3
11.3
0.0
31.3
11.6
20.0
31.3
0.0
Pass
Pass
Mean |Difference| in Reduction (%)
0
5
10
15
20
25
30
Pass Co-benefit Matrix
(fraction of circuits where both passes achieve > 0 reduction)
1.00
0.00
0.67
0.33
0.00
0.00
0.00
0.00
0.67
0.00
1.00
0.00
0.33
0.00
0.00
1.00
Co-benefit Fraction
0.0
0.2
0.4
0.6
0.8
1.0